

APEX INSTITUTE OF MEDICAL SCIENCES & RESEARCH

Department of Clinical Microbiology & Infectious Diseases

NABH & NABL ACCREDITED

BLOOD CULTURE & AUTOMATED ANTIBIOGRAM REPORT

REF: BC-2026-98241

Patient ID / Alias: PT-1042 (ICU Bed 08) **Referring Dept:** Critical Care Medicine **Collection Date:** 11-Sep-2026 08:30 AM
Age / Gender: 62 Y / Male **Allergy Status:** Childhood rash to Amoxicillin (No severe reactions) **Site:** Peripheral Venous Catheter (Port C)
Location: Intensive Care Unit (ICU) **Current Antibiotic:** Meropenem 1g IV TDS (Empiric Broad-Spectrum)

SPECIMEN & MICROBIOLOGICAL IDENTIFICATION

Specimen Type: Blood (Peripheral Venipuncture) **Isolated Organism:** *Escherichia coli* (>10⁵ CFU/mL)

Direct Smear: Gram-negative bacilli observed in aerobic blood culture. **Colony Count:** 1.2 x 10⁸ CFU/mL. **Tested:** Bactec 9120 positive flag

ANTIMICROBIAL SUSCEPTIBILITY TESTING (CLSI M100 / EUCAST CRITERIA)

ANTIMICROBIAL AGENT	WHO AWARE	MIC (ug/mL)	INTERPRETATION
Ceftriaxone	Watch Tier	<=0.5	SUSCEPTIBLE
Meropenem	Watch Tier	<=0.25	SUSCEPTIBLE
Piperacillin-Tazobactam	Watch Tier	<=8/4	SUSCEPTIBLE
Amikacin	Access Tier	<=4.0	SUSCEPTIBLE
Amoxicillin-Clavulanate	Access Tier	>=32/16	RESISTANT
Ciprofloxacin	Watch Tier	>=4.0	RESISTANT

BIOCHEMICAL & ORGAN FUNCTION BIOMARKERS

Serum Creatinine	eGFR Clearance	Total Leukocyte (TLC)	Procalcitonin
1.8 mg/dL (Elevated)	38 mL/min/1.73m ²	14,200 /uL	2.4 ng/mL

ANTIMICROBIAL STEWARDSHIP CLINICAL SUMMARY & ACTION PLAN

TARGETED RECOMMENDATION: Ceftriaxone 2g IV Once Daily (OD) for 7 to 10 days.

Clinical Rationale: Definitive blood culture confirms *E. coli* susceptible to Ceftriaxone. De-escalate from Meropenem to preserve Reserve/Watchlist agents.

Safety & Monitoring: Serum creatinine is elevated (1.8 mg/dL, tested 72h ago). Draw repeat renal panel within 48 hours. Verify patient remains afebrile.

Stewardship & Economic Impact: Carbapenem-sparing protocol saves estimated ₹3,140/day in medication cost and reduces risk of C. difficile.